

NANYANG JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATION  
Higher 2

CANDIDATE  
NAME

CLASS

TUTOR'S  
NAME

## PHYSICS

**9749/03**

Paper 3 Longer Structured Questions

**20 September 2018**

**2 hours**

Candidates answer on the Question Paper.

No Additional Materials are required

### READ THESE INSTRUCTIONS FIRST

Write your name, class, Centre number and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

#### Section A

Answer **all** questions.

#### Section B

Answer any **one** question.

You are advised to spend about one and a half hours on Section A, and about thirty minutes on Section B.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use |      |
|--------------------|------|
| Section A          |      |
| 1                  | / 7  |
| 2                  | / 6  |
| 3                  | / 8  |
| 4                  | / 10 |
| 5                  | / 10 |
| 6                  | / 7  |
| 7                  | / 12 |
| Section B          |      |
| 8                  | / 20 |
| 9                  | / 20 |
| Total              |      |

This document consists of **21** printed pages.

**Data**

speed of light in free space

$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$

permeability of free space

$$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$$

permittivity of free space

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1} \\ (1 / (36\pi)) \times 10^{-9} \text{ F m}^{-1}$$

elementary charge

$$e = 1.60 \times 10^{-19} \text{ C}$$

the Planck constant

$$h = 6.63 \times 10^{-34} \text{ J s}$$

unified atomic mass constant

$$u = 1.66 \times 10^{-27} \text{ kg}$$

rest mass of electron

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

rest mass of proton

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

molar gas constant

$$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

the Avogadro constant

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

the Boltzmann constant

$$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

gravitational constant,

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

acceleration of free fall

$$g = 9.81 \text{ m s}^{-2}$$

**Formulae**

uniformly accelerated motion

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas

$$W = p\Delta V$$

hydrostatic pressure

$$p = \rho gh$$

gravitational potential

$$\phi = -Gm / r$$

temperature

$$T/\text{K} = T/^{\circ}\text{C} + 273.15$$

pressure of an ideal gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

mean translational kinetic energy of an ideal molecule

$$E = \frac{3}{2} kT$$

displacement of particle in s.h.m.

$$x = x_0 \sin \omega t$$

velocity of particle in s.h.m.

$$v = v_0 \cos \omega t \\ = \pm \omega \sqrt{(x_0^2 - x^2)}$$

electric current

$$I = Anvq$$

resistors in series

$$R = R_1 + R_2 + \dots$$

resistors in parallel

$$1/R = 1/R_1 + 1/R_2 + \dots$$

electric potential

$$V = Q / 4\pi\epsilon_0 r$$

alternating current/voltage

$$x = x_0 \sin \omega t$$

magnetic flux density due to a long straight wire

$$B = \mu_0 I / 2\pi d$$

magnetic flux density due to a flat circular coil

$$B = \mu_0 NI / 2r$$

magnetic flux density due to a long solenoid

$$B = \mu_0 nI$$

radioactive decay

$$x = x_0 \exp(-\lambda t)$$

decay constant

$$\lambda = \ln 2 / t_{1/2}$$

## Section A

Answer **all** the questions in the spaces provided.

- 1** Capacitance measures the ability of an electrical component to store charge and is defined as the ratio of charge stored in a capacitor to the potential difference across it.

**(a)** The S.I. unit for capacitance is the farad (F). Express F in S.I. base units.

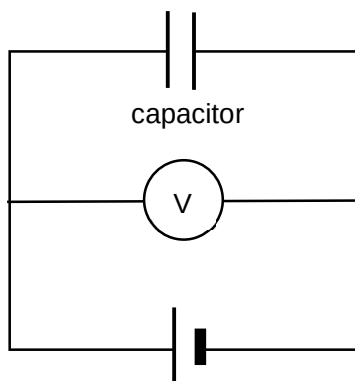
$$F = C V^{-1}$$

$$C = A s \text{ and } V = [W/q] = kg \, m^2 \, s^{-2} \, A^{-1} \, s^{-1} = kg \, m^2 \, s^{-3} \, A^{-1}$$

$$F = kg^{-1} \, m^{-2} \, s^4 \, A^2$$

$$F = \dots\dots\dots [3]$$

**(b)** A capacitor of capacitance  $50 \, \mu F$  is charged using a cell in the circuit shown in Fig. 1.1.



**Fig. 1.1**

- (i)** The voltmeter reads  $2.20 \, V$ . By applying the given definition of capacitance, calculate the charge stored in the capacitor to 3 significant figures.

$$C = Q / V \rightarrow Q = 50 \times 10^{-6} \times 2.20 = 1.10 \times 10^{-4} \, C$$

- (ii)** The uncertainty of the voltmeter reading is  $\pm 0.05 \, V$ , and the capacitance of the capacitor has a tolerance of  $\pm 10\%$ .

Express the charge stored in the capacitor in the form of  $(Q \pm \Delta Q)$ .

$$Q = C V$$

$$\Delta Q/Q = \Delta C/C + \Delta V/V = 0.10 + 0.05/2.20 = 0.12$$

$$\Delta Q = 0.1 \times 10^{-4} \, C$$

$$Q \pm \Delta Q = (1.1 \pm 0.1) \times 10^{-4} \, C$$

- 2** A person supports a load of  $20 \, N$  in his hand as shown in Fig. 2.1. The system consisting of